

Digital Forensics



File System Overview



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File System Overview



- A file system stores data on a device so data can be retrieved by the system or a user.
- File systems are largely independent of an operating system (OS), and different file systems can be supported on different OSs when necessary drivers are installed.
- File systems are formatted onto a volume – either a single-partition or a multiple-partition volume in the case of a RAID or an LVM.



File System Overview



- The file system provides a means by which the OS can access the actual files stored on disk.
- At its core, a file system can be thought of as a database in the sense that, like a database, data is stored and metadata is used to track the data.
- Modern file systems are largely independent of the OS – Linux can read and write to the Microsoft NTFS, and with the proper drivers, Windows can read and write to Linux EXT (Extended File System).



File System Overview



- Metadata in a file system includes items like a file's location (or address) and the size of the data so the system can retrieve it from the hardware.
- A file system will track the location of a file's data location as block addresses.
- Multiple blocks of data may be necessary for files that are too large to be stored in a single block and/or cannot be stored in consecutive blocks on the device.



File System Overview



- **General File System Information:** i.e. general structure of file system, location of certain data structures and size of a data unit
- **Content of the files:** Which is generally organized into a collection of standard sized containers like clusters, blocks.
- **Meta data of files:** FAT directory entries, NTFS MFT entries, UFS and Ext3 inode structures are examples of this type of data structure.
- **File name**
- **User quota statistics and**
- **file system journals.**



File System Overview



- Metadata also includes items like a file name and the location of a file in hierarchical directory structures or paths so the user can find the file.
- File names can be limited like in the FAT16 file system to eight bytes with a three-byte file extension, or they can be long filenames such as is the case with most modern file systems.
- Of course, even filenames have limitations that can be limited by the file system, or in some cases limited by the OS and the ability to address finite characters in a file's full path.



File System Overview



- Microsoft file systems utilize clusters in addressing, while Linux nomenclature refers to addressing in blocks.
- A cluster or a block consists of one or more sectors. Formatting options in file systems allow the user to customize the cluster size or allow the OS to automatically customize this.
- A typical Windows system running an NTFS file system on a one-terabyte drive will likely contain a cluster size equal to 4096 bytes – or eight sectors per cluster.



File System Overview



- This allows for larger drives to be utilized because the file system only has to address one-eighth the number of sectors.
- The impact of this, then, is that if a one-byte file is created, it will occupy 4096 bytes – or eight sectors – of space.



File System Overview



- Essential file system data are those that are needed to save and retrieve files.
- Examples of this type of data include the addresses where the file content is stored, the name of a file, and the pointer from a name to a metadata structure.
- Non-essential file system data are those that are there for convenience but not needed for the basic functionality of saving and retrieving files.
- Access times and permissions are examples of this type of data.



File System Overview



- NTFS
- FAT File System
- Unix File System (Ext, Ext2, Ext3, Ext4)
- HFS+, APFS
- UFS, CDFS



References



- Digital Forensics by André Årnes
- Computer Systems: Digital Design, Fundamentals of Computer Architecture and Assembly Language by Ata Elahi
- Practical Forensic Imaging: Securing Digital Evidence with Linux Tools by Bruce Nikkel
- A Practical Guide to Digital Forensics Investigations 2nd Edition by Darren R. Hayes

